EXAM SHIELD

DevOps Engineering Question Bank

Containers, orchestration, automation and observability.

13 questions: 8 multiple choice, 3 fill in the gap and 2 written. Total 20 marks.
Edition: with answers (staff copy, ready to import into Exam Shield).

How this paper is laid out

Questions are numbered in one sequence. Multiple-choice options are lettered with a full stop; each multiple-choice question is worth one mark unless stated.

Fill-in-the-gap questions show each blank as a line; they earn one mark per blank, shown in square brackets, and are marked automatically.

Written questions show their marks in square brackets and are marked by an examiner.

To use this paper in Exam Shield, open an exam, choose Questions and then Import from PDF, and upload this file. Review the preview and import.

Section 1: DevOps Engineering

Containers, orchestration, automation and observability.

1. In Docker, what is a container?

A. A hypervisor for virtual machines
B. A registry that stores images
C. A running instance of an image
D. A script that builds an image

Answer: C

2. Which file describes how to build a Docker image?

A. Dockerfile
B. Makefile
C. Jenkinsfile
D. package.json

Answer: A

3. What is the smallest deployable unit in Kubernetes?

A. Cluster
B. Pod
C. Namespace
D. Node

Answer: B

4. Which tool is used for Infrastructure as Code?

A. Terraform
B. Wireshark
C. Postman
D. Photoshop

Answer: A

5. What is a blue-green deployment?

A. A debugging technique
B. Running two identical environments and switching traffic between them
C. A code formatting standard
D. A colour scheme for dashboards

Answer: B

6. Which tool collects and stores time-series metrics for monitoring?

A. Webpack
B. Babel
C. Prometheus
D. Figma

Answer: C

7. In a CI/CD pipeline, what normally happens in the test stage?

A. Servers are physically installed
B. Automated tests run against the new build
C. Old builds are archived
D. The release is announced to customers

Answer: B

8. Which command lists the running Docker containers?

A. docker show
B. docker ps
C. docker ls-run
D. docker images

Answer: B

9. Releasing a new version to a small share of users before everyone else is a _____ deployment. [1 mark]

Answer: canary

10. Ansible, Puppet and Chef are _____ management tools. [1 mark]

Answer: configuration

11. Keeping the desired state of infrastructure in Git and applying changes automatically from it is called _____. [1 mark]

Answer: GitOps

12. Explain what a CI/CD pipeline is and describe three benefits it brings. [5 marks]

13. Compare containers with virtual machines in terms of isolation, size and start-up time. [4 marks]